

Technology Activities and Cognitive Trajectories Among Community-Dwelling Older Adults

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Introduction:

- While prior research has established a relationship between digital technology use and improved cognitive function among older adults, the impacts of different technology activities on specific cognitive domains remain underexplored.
- Research questions include
 - What are the associations between technology activities and cognitive domains for community-dwelling older adults without dementia?
 - Do the start and stop of using specific technology activities moderate the rate of cognitive decline over time in community-dwelling older adults without dementia?

Method:

- The sample included 5,566 community-dwelling older adults without dementia from the **National Health and Aging Trends Study (NHATS)** (2015-2022).
- Technology activities included online shopping, banking, medication refills, social network site visits, and checking health conditions online.
- The cognitive domains assessed were episodic memory (0-20), executive function (0-5), and orientation (0-8), all standardized to Z-scores.
- Covariates included sex, age, race, education, living arrangement, income, daily and instrumental activity difficulties, self-rated health, rurality, and device ownership.
- GLMM and within-person asymmetric effect were used.
- Benjamin-Hochberg false discovery rate (FDR) correction was applied to reduce the risk of type I errors.
- 1-year lagged models were applied for sensitivity analysis.

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Initiating and maintaining engagement with specific digital activities – such as medication refills, online banking, shopping, and social media – are modestly associated with slower cognitive decline in older adults, particularly in episodic memory and orientation.

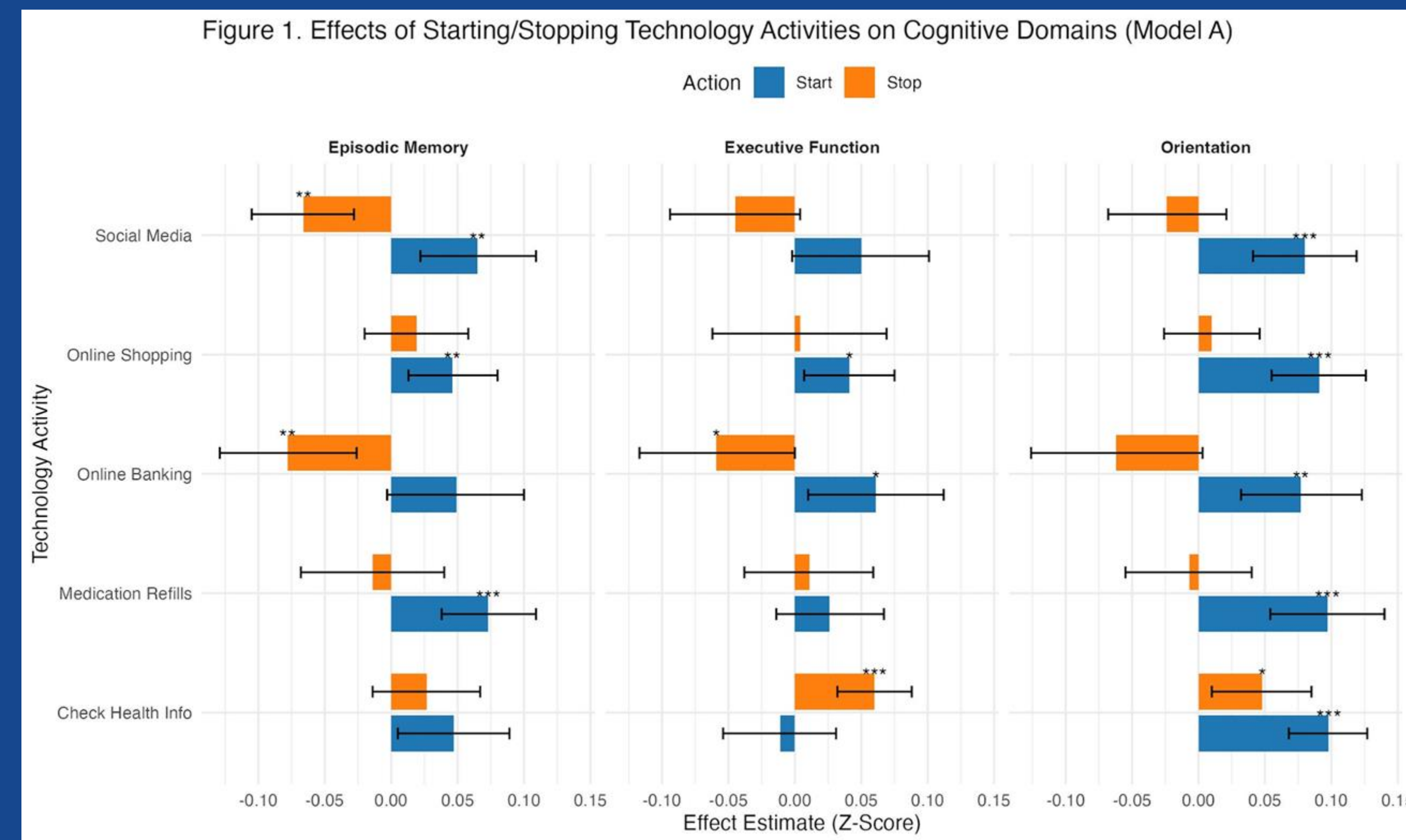


Table 1. The Effects of Start/Stop Technology Activities on Cognitive Domains (Model A)						
	Episodic Memory Z-Score		Executive Function Z-Score		Orientation Z-Score	
	Start	Stop	Start	Stop	Start	Stop
Social Media	0.065** [0.022,0.109]	-0.066** [-0.105,-0.028]	0.050 [0.002,0.101]	-0.045 [-0.094,0.004]	0.080*** [0.041,0.119]	-0.024 [-0.068,0.021]
Online Shopping	0.046** [0.013,0.080]	0.019 [-0.020,0.058]	0.041* [0.007,0.075]	0.004 [-0.062,0.069]	0.091*** [0.055,0.126]	0.010 [-0.026,0.046]
Online Banking	0.049 [-0.003,0.100]	-0.078** [-0.129,-0.026]	0.061* [0.010,0.112]	-0.059* [-0.117,-0.000]	0.077** [0.032,0.123]	-0.062 [-0.126,0.003]
Medication Refills	0.073*** [0.038,0.109]	-0.014 [-0.068,0.040]	0.026 [-0.014,0.067]	0.011 [-0.038,0.059]	0.097*** [0.054,0.140]	-0.007 [-0.055,0.040]
Check Health Info	0.047 [0.005,0.089]	0.027 [-0.014,0.067]	-0.011 [-0.054,0.031]	0.060*** [0.032,0.088]	0.098*** [0.068,0.127]	0.048* [0.010,0.085]
Note: p-values are marked after false discovery rate (FDR) using Benjamini-Hochberg method 95% confidence intervals in brackets *p<0.05, **p<0.01, ***p<0.001						

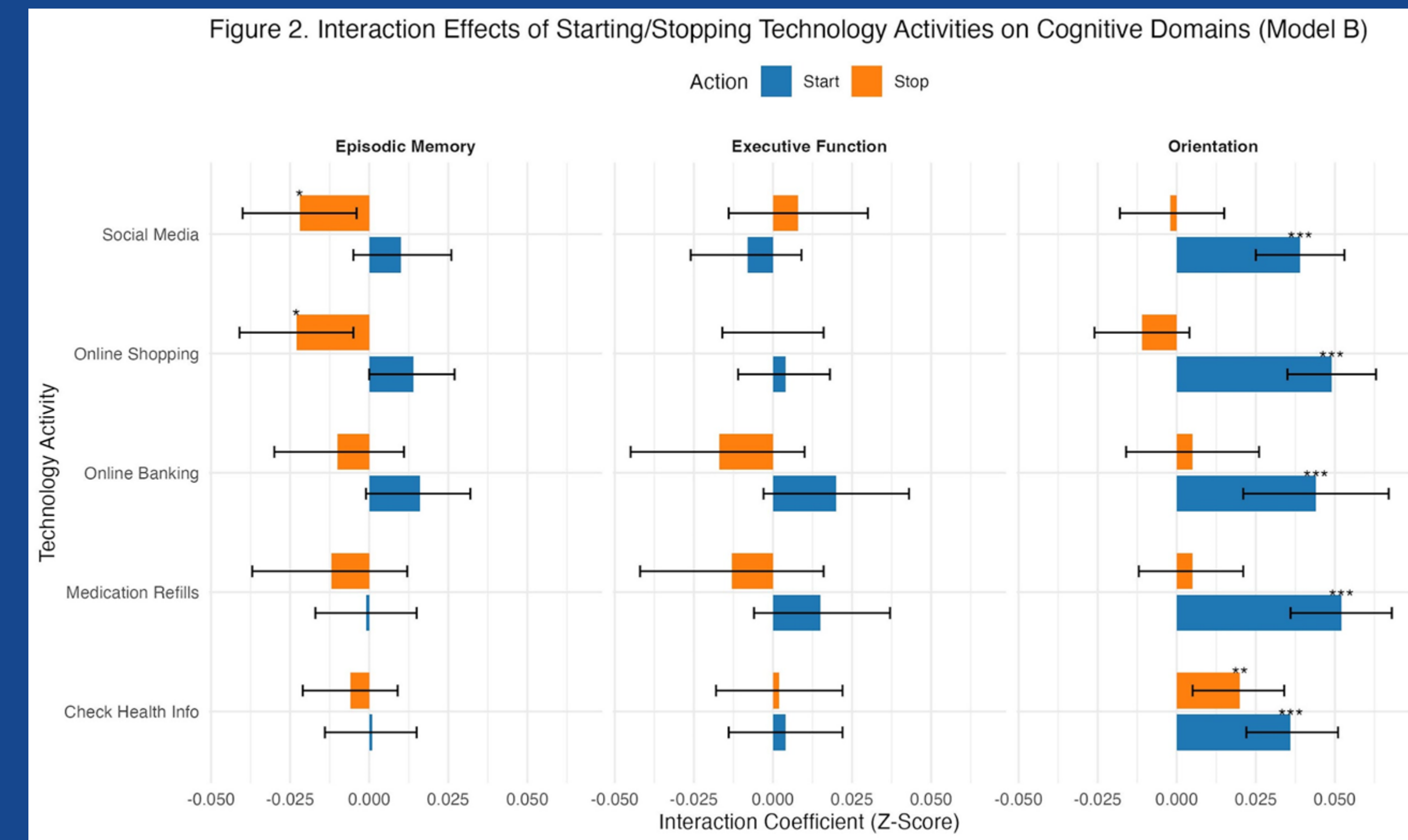


Table 2. The Interaction Coefficients of Start/Stop Technology Activities on Cognitive Domains (Model B)						
	Episodic Memory Z-Score		Executive Function Z-Score		Orientation Z-Score	
	Start	Stop	Start	Stop	Start	Stop
Social Media	0.010 [-0.005,0.026]	-0.022 [-0.040,-0.004]	-0.008 [-0.026,0.009]	0.008 [-0.014,0.030]	0.039*** [0.025,0.053]	-0.002 [-0.018,0.015]
Online Shopping	0.014 [0.000,0.027]	-0.023* [-0.041,-0.005]	0.004 [-0.011,0.018]	-0.000 [-0.016,0.016]	0.049*** [0.035,0.063]	-0.011 [-0.026,0.004]
Online Banking	0.016 [-0.001,0.032]	-0.010 [-0.030,0.011]	0.020 [-0.003,0.043]	-0.017 [-0.045,0.010]	0.044*** [0.021,0.067]	0.005 [-0.016,0.026]
Medication Refills	-0.001 [-0.017,0.015]	-0.012 [-0.037,0.012]	0.015 [-0.006,0.037]	-0.013 [-0.042,0.016]	0.052*** [0.036,0.068]	0.005 [-0.012,0.021]
Check Health Info	0.001 [-0.014,0.015]	-0.006 [-0.021,0.009]	0.004 [-0.014,0.022]	0.002 [-0.018,0.022]	0.036*** [0.022,0.051]	0.020** [0.005,0.034]
Note: p-values are marked after false discovery rate (FDR) using Benjamini-Hochberg method 95% confidence intervals in brackets *p<0.05, **p<0.01, ***p<0.001						

Results:

- Onset of digital activities** was associated with **higher episodic memory score**
 - Online shopping ($\beta = 0.046^{**}$)
 - Medication refills ($\beta = 0.073^{***}$)
 - Social media use ($\beta = 0.065^{**}$)
- Cessation of digital activities** was associated with **lower episodic memory score**
 - Online banking ($\beta = -0.078^{**}$)
 - Social media use ($\beta = -0.066^{**}$)
- Stopping online shopping** was associated with a **slower cognitive decline in episodic memory** ($\beta = -0.023^{*}$)
- Initiating instrumental, social, and health-related digital activities was associated with slower cognitive decline in orientation.
- Lagged models highlighted
 - Negative impact of stopping online banking on episodic memory
 - Positive effects of starting online medication refills
 - Online shopping and social media use linked to better orientation
- All observed effects were statistically significant but small in magnitude

Conclusions:

- Promoting **online medication refill use** and **maintaining online banking** may help support episodic memory in older adults.
- To support **orientation**, encourage the **adoption of online shopping and social media** platforms—especially for those unfamiliar or newly engaging.
- Future trials should:
 - Evaluate the **clinical relevance** of targeted digital interventions
 - Promote **initiation and sustainment** of digital activities
 - Aim to mitigate **domain-specific cognitive decline** in aging populations

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